How Do Users Review Spotify? Analyzing Emotion, Topics, and Platform Replies Using Text-as-Data

Yihan Chen[a,1]

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This study analyzes 60,000 Spotify reviews to examine what drives platform replies. Using text classification, topic modeling, and logistic regression, we find replies are more likely for detailed, negative feedback—especially with technical issues or emotions like disgust. Joy, praise, and competitor mentions have little impact.

Spotify | User Reviews | Text Analysis | Platform Prioritization | Sentiment Analysis

Spotify has become the global leader in music streaming, with over 252 million premium subscribers as of Q3 2024 (Statista, 2024; Figure 1).

While subscription metrics reveal growth, understanding user satisfaction and retention requires deeper analysis beyond numerical figures. User-generated content (UGC), especially app store reviews, provides insights into customer experiences.

These reviews reflect not only product satisfaction but also user frustrations and unmet expectations. Platforms like Spotify occasionally respond to user reviews, but the dynamics of which reviews receive replies remain largely unexplored.

1. Introduction

1.1 Motivation. From a business strategy perspective, responding to user feedback can enhance loyalty, mitigate churn risks, and signal active customer engagement. From a policy perspective, increasing attention is being paid to platform transparency, consumer protection, and fairness in digital services (OECD, 2022).

Understanding how Spotify engages with user feedback through public responses has important implications for platform governance, user retention strategies, and broader digital platform accountability.

1.2 Prior Research. Literature supports that text analysis informs consumer insights. Pang Lee (2008), Liu (2012) made the foundation of sentiment analysis for understanding customer satisfaction. Tirunillai Tellis (2012) showed that user-generated content predicts business performance, while Blei et al. (2003) introduced LDA topic modeling to identify consumer concerns. In digital media, Datta et al. (2018) found that users' sentiments correlate with music subscriber growth, and Piskorski et al. (2019) demonstrated that negative sentiment spikes signal churn risk. These findings help to support the analysis of user sentiment in Spotify's reviews and make market decisions.

1.3 Research Questions. RQ1: How do Spotify users describe their experience in written reviews?
RQ2: Which review topics or tones are more likely to receive an official reply—and what does this reveal about platform priorities or transparency?

1.4 Report Overview. The report describes the dataset and annotation, followed by supervised classification and topic modeling. I then evaluate emotional tone and reply likelihood through logistic regression, concluding with discussion.

2. Data and Methods

2.1 Data Source and Unit of Observation. The dataset, published on Kaggle (2022), contains 61,594 Spotify app reviews from the Google Play Store. Each entry includes the text of the review, timestamp, rating (1–5), helpfulness score, and reply (if any). Each observation corresponds to one user's review.

Significance Statement

Understanding how Spotify selectively replies to user reviews sheds light on platform governance, digital fairness, and customer engagement strategy. By analyzing emotional tone and review content using TAD and logistic regression, this study highlights patterns in review topics and corporate responsiveness, contributeing to business insight and broader discussions on digital transparency and priority.

Author affiliations: [a]Georgetown University, McCourt School of Public Policy, Washington, DC 20001

[1]To whom correspondence should be addressed. E-mail: yc1140@georgetown.edu

2.2 Variables of Interest. The dataset used in this study comprises a set of variables that support both classification and regression analysis. (Table 1) Each review is accompanied by structured metadata and derived features that inform our analysis of user experience and reply behavior.

2.3 Data Wrangling Steps. To prepare the dataset for analysis, I implemented a structured data cleaning and transformation pipeline.

First, I converted all review text to lowercase, removed duplicates, and eliminated special characters to ensure consistency. The cleaned text was then tokenized and matched with the NRC lexicon to compute review-level emotion scores across ten emotion categories.

Using a manually labeled random sample of 2,000 reviews, I initially created four functional categories: Technical Issues, UX Features, Price Ads, and Other. However, due to overlapping themes and classification ambiguity—especially between Technical and UX feedback—I consolidated these into a binary scheme to improve annotation consistency and model performance. Therefore, the final categories were: (1) “App Experience”: reviews mainly focused on technical problems, user interface, and app functionality; (2) “Other”: reviews related to pricing, ads, music content, or general sentiment. The ”App Experience” category may include all forms of interaction with the Spotify platform, including the mobile app, website, and other interfaces. It may also cover reviews that partially mention pricing, praise, or subscription but are primarily focused on user experience. This simplified structure allowed the classifier to more reliably distinguish actionable user feedback tied to user experience and these two labeled subsets was used for training supervised classification models.

I also engineered several features to support downstream analysis. Binary flags were created to indicate whether a review received an official Spotify reply and whether a competing platform (e.g., Apple Music) was mentioned. I computed word counts for each review and detected specific competitor references.

For classification, I constructed both TF-IDF and word embedding matrices to predict topic labels. To perform unsupervised topic modeling, I filtered the App Experience reviews and preprocessed them for input into an LDA model.

To address class imbalance in reply behavior, I applied SMOTE (Synthetic Minority Oversampling Technique), generating synthetic samples for the minority class (Replied = 1). This rebalancing allowed me to fit more robust logistic regression models.

3. Analysis

3.1 Supervised Classification. To distinguish between App Experience and Other reviews, I trained three supervised models—Naïve Bayes, SVM, and Lasso—on the annotated subset. Among these, the Lasso logistic regression model achieved the highest accuracy (83%) and interpretability (Table 2). I applied this model to classify the remaining reviews in the dataset.

3.2 Topic Modeling. LDA was applied to the App Experience subset ($n \approx 38{,}000$) to identify six subthemes (Table 3): Technical Fixes, Account Subscription, General UX Feedback, Playback Issues, Music Playlist Features, and Positive Experience.

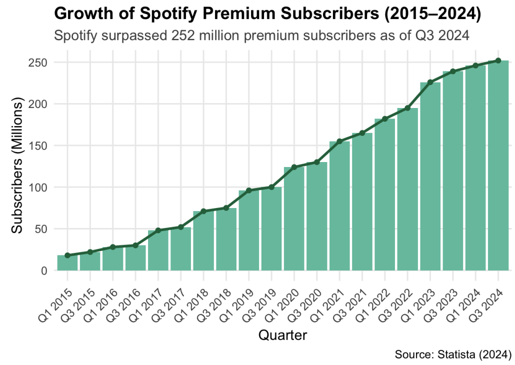

Fig. 1. Number of Spotify premium subscribers worldwide from Q1 2015 to Q4 2024 (Source: Statista, 2024).

Table 1. Variables of Interest in the Spotify Review Dataset

Variable Name	Description
Review	Full text of the user review
Rating	Star rating (1 to 5 scale)
Predicted_Label	Binary label (App Experience vs. Other) via TF-IDF Lasso
Sentiment	Derived from rating: Negative (1–2), Neutral (3), Positive (4–5)
Replied	Binary indicator if Spotify replied
PredictedLabel_LDA	Topic label from LDA topic modeling
NRC Emotion	Ten emotion scores from NRC Lexicon
word_count	Total number of words in review
competitor_mention	Binary indicator if a competitor is mentioned

Table 2. Performance Metrics of Lasso Models

Metric	TF-IDF vs. Word Embedding
Accuracy	0.83, 0.82
Kappa	0.648, 0.613
Precision	0.868, 0.837
Recall	0.846, 0.878
F1 Score	0.857, 0.857
Balanced Accuracy	0.826, 0.802

Table 3. LDA-Inferred Topics for App Experience Reviews

Topic	Interpretation
Technical Fixes	Bugs, app malfunctions, connection issues
Account & Subscription	Login, billing, and premium access issues
General UX Feedback	Vague usability comments and minor frustrations
Playback Issues	Playback interruptions, especially after updates
Music and Playlist Features	Playlist quality, song selection, and feature feedback
Positive Experience	Expressions of satisfaction and brand loyalty

Fig. 2. Descriptive overview: sentiment tone and App Experience classification share among Spotify reviews.

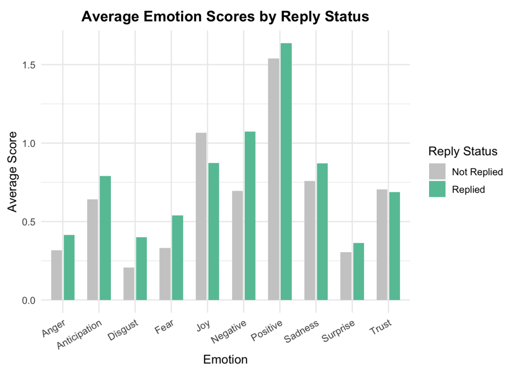

Fig. 3. Average emotion scores by reply status: replied vs. unreplied reviews across ten NRC emotion categories.

3.3 Sentiment Analysis. Sentiment was categorized based on ratings: 1–2 as negative, 3 neutral, and 4–5 positive. NRC Lexicon provided 10 emotion scores (e.g., anger, joy, disgust) for each review.

3.4 Logistic Regression: Reply Prediction. I modeled reply likelihood using emotional scores, review length, competitor mentions, and topic labels. A second model applied SMOTE to account for class imbalance (Figure 7).

4. Results

4.1 Descriptive Overview. 49 percent of Spotify reviews are positive in tone, with around 60 percent classified under the App Experience category. However, only 3.5 percent of reviews received an official reply from the platform, highlighting a significant gap in engagement (Figure 2).

4.2 Sentiment, Emotion Score, and Reply. Spotify tends not to respond to emotionally extreme or overtly positive reviews; instead, it is more likely to engage with detailed expressions of dissatisfaction. Reviews expressing disgust or positive sentiment are more likely to receive a response (Figure 3), with disgust showing a significant positive effect (coefficient = 0.464, $p < 0.001$). In contrast, reviews characterized by anger, joy, or trust tend to be overlooked—joy (coefficient = -0.240, $p < 0.05$) and trust (coefficient = -0.226, $p < 0.05$) both significantly reduce the likelihood of a reply.

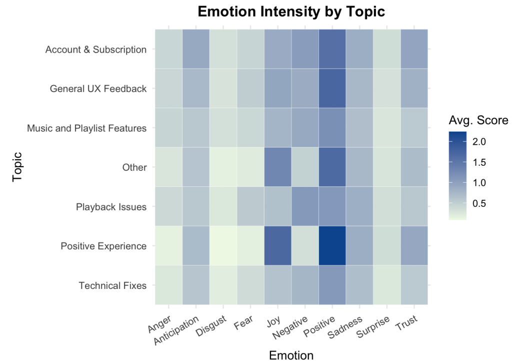

Fig. 4. Emotion scores by topic: each LDA-derived topic displays its corresponding average emotion levels using the NRC lexicon.

4.3 Topics: Spotify Prioritizes Problem-Solving. Spotify appears more responsive to critical reviews tied to app performance than to general praise. Approximately 51% of reviews classified as App Experience were negative, compared to just 23% in the Other category. Furthermore, over 75% of all replies were directed to App Experience reviews.

Different topics share broadly similar distributions of emotion scores (Figure 4), with one key exception: the Positive Experience topic tends to reflect high levels of joy and positive sentiment, yet is one of the least likely to receive a reply.

Spotify prioritizes resolving technical access and service issues over addressing feature feedback or acknowledging general satisfaction. Compared to the reference category (Account & Subscription), reviews classified as Music and Playlist Features (coefficient = -0.717, $p < 0.05$), Other (coefficient = -1.031, $p < 0.001$), and Positive Experience (coefficient = -0.915, $p < 0.01$) are significantly less likely to receive a response.

4.4 Competitor Mentions: Mentioning Others Doesn't Get Noticed. The ten most frequently mentioned competitors in Spotify reviews are Pandora, YouTube Music, Apple Music, Amazon Music, SoundCloud, Tidal, Deezer, Gaana, Wynk, and Napster. Among the top 5 mentioned competitors, Apple Music, YouTube Music, and Amazon Music tend to show the most joy, positive sentiment, and sadness scores (Figure 5). SoundCloud, YouTube Music, and Amazon Music are often mentioned in negative contexts, whereas Apple Music and Pandora are more frequently cited in positive or neutral reviews).

However, Spotify did not explicitly engage with replies containing competitors. According to the logistic regression model, the presence of a competitor mention is not a statistically significant predictor of receiving a reply (coefficient = -0.103, $p > 0.05$).

4.5 Word Count: Spotify Values Detailed User Feedback. Most reviews are relatively concise: the average review contains 32 words, and the interquartile range spans from 14 to 43 words, though some extend up to 738. Reviews that are longer in length are significantly more likely to receive a response: each additional word increases the probability of receiving a reply by approximately 1% (coefficient = 0.010, $p < 0.01$).

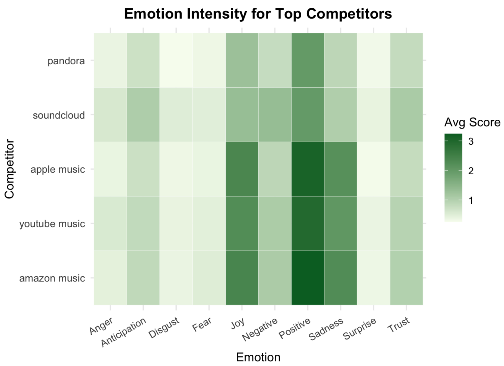

Fig. 5. Emotion scores by top 5 competitors

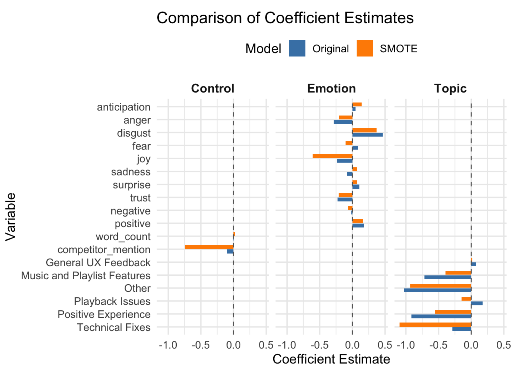

Fig. 6. Comparison of Coefficient Estimates

4.6 SMOTE Adjustment for Class Imbalance. Spotify's engagement is systematically focused on complaints, even under balanced sampling. To validate the robustness of these findings, I re-estimated the logistic regression using a SMOTE-balanced dataset to address the low reply rate. The results were largely consistent: emotionally negative tones like disgust and technical topics like Account & Subscription remained strong predictors of reply likelihood (Table 4; (Figure 6)).

5. Discussion

Spotify replies to only a small fraction of user reviews and does so selectively, prioritizing longer, more detailed reviews that express dissatisfaction, especially those involving technical or service-related issues. Reviews with emotionally nuanced tones like disgust are more likely to receive a response, while those expressing joy, trust, or general praise tend to be overlooked. Topic modeling and regression results further confirm that Spotify is most responsive to reviews under categories like Account Subscription. Mentioning competitors such as Apple Music or YouTube Music does not significantly influence reply behavior.

From a business strategy perspective, the analysis identifies key experience themes and sentiment patterns that can inform improvements in Spotify's app functionality, user satisfaction, and retention efforts. By highlighting which types of reviews prompt engagement—namely those that are detailed, emotionally nuanced, and related to technical service—Spotify can better prioritize customer service efforts and refine its product experience.

From a governance perspective, this study provides transparency into how Spotify manages public engagement through app store replies. It reveals selective responsiveness to certain emotional tones and topics, raising broader questions about fairness, accountability, and communication standards in platform behavior. The use of text-as-data methods contributes to ongoing policy discussions by empirically examining how platform design choices affect user interaction and perceived responsiveness.

Future research could expand this analysis by incorporating data from other platforms (e.g., Apple or YouTube Music) to allow for comparative studies. Further improvements could also include training deep learning models on a larger, more diverse dataset or integrating user metadata to explore demographic patterns in reply behavior. Besides, while reply patterns suggest engagement priorities, further work is needed to fully assess transparency and policy implications.

5.1 Limitations. While the findings offer clear patterns, several limitations remain. First, TF-IDF outperformed word embeddings in classification, which may reflect insufficient embedding tuning or the brevity of most reviews.

Second, topic modeling and classification outputs showed some conceptual overlap, limiting interpretability. Enhanced preprocessing or alternative models could improve topic separation.

Third, although the number of retained duplicate reviews is small (270), their short and repetitive nature may reduce analytical quality. Word clouds also revealed minor stemming issues, such as splitting "song" and "songs."

Finally, fewer than 3.5% of reviews received a reply, significantly limiting the statistical power of the regression model.

R Packages Used. This project relied on a range of R packages, organized by functionality:

Data Cleaning and Manipulation: `tidyverse`, `janitor`, `textclean`, and `hunspell` were used for cleaning and transforming raw text, handling missing data, and standardizing review content.

Text Preprocessing and Feature Engineering: `tidytext`, `quanteda`, `quanteda.textstats`, and `textmineR` facilitated tokenization, DTM construction, word frequency analysis, and lexicon-based scoring.

Topic Modeling: For unsupervised topic extraction, I used `topicmodels` (LDA) along with `broom.mixed` for tidy outputs.

Supervised Classification: I implemented `caret`, `naivebayes`, `glmnet`, and `e1071` to build and evaluate machine learning models (e.g., Lasso, SVM, Naive Bayes) for predicting review types.

Visualization: `ggplot2`, `scales`, `patchwork`, `wordcloud`, and `RColorBrewer` were used to create comparative bar plots, emotion visualizations, and topic word clouds.

Model Interpretation and Reporting: `pscl`, `modelsummary`, and `gridExtra` supported logistic regression analysis and presentation of results.

Class Imbalance Handling: `smotefamily` was used to apply the SMOTE algorithm for rebalancing the reply classification task.

All packages were managed using `pacman::p_load()` for efficient loading and installation.

$$
\begin{aligned}
\text{logit}(\Pr(\text{Replied}_i = 1)) = \; & \beta_0 + \beta_1 \cdot \text{anticipation}_i + \beta_2 \cdot \text{anger}_i + \beta_3 \cdot \text{disgust}_i + \beta_4 \cdot \text{fear}_i \\
& + \beta_5 \cdot \text{joy}_i + \beta_6 \cdot \text{sadness}_i + \beta_7 \cdot \text{surprise}_i + \beta_8 \cdot \text{trust}_i \\
& + \beta_9 \cdot \text{negative}_i + \beta_{10} \cdot \text{positive}_i + \beta_{11} \cdot \text{word\_count}_i \\
& + \beta_{12} \cdot \text{competitor\_mention}_i + \beta_{13} \cdot \text{PredictedLabel\_LDA}_i + \varepsilon_i
\end{aligned} \quad [1]
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Fig. 7. Logistic regression model specification for predicting reply likelihood.

Table 4. Logistic Regression: Predicting Spotify Replies (Original vs. SMOTE Adjusted)

Variable	Category	Original Model	SMOTE Adjusted Model
(Intercept)	—	-5.528^{***} (0.207)	0.011 (0.021)
anticipation	NRC Lexicon	0.049 (0.090)	0.142^{***} (0.011)
anger	NRC Lexicon	-0.283^{*} (0.137)	-0.202^{***} (0.016)
disgust	NRC Lexicon	0.464^{***} (0.138)	0.371^{***} (0.017)
fear	NRC Lexicon	0.086 (0.119)	-0.106^{***} (0.014)
joy	NRC Lexicon	-0.240^{*} (0.121)	-0.609^{***} (0.014)
sadness	NRC Lexicon	-0.079 (0.107)	0.069^{***} (0.012)
surprise	NRC Lexicon	0.107 (0.130)	0.070^{***} (0.014)
trust	NRC Lexicon	-0.226^{*} (0.105)	-0.212^{***} (0.011)
negative	NRC Lexicon	-0.017 (0.096)	-0.062^{***} (0.011)
positive	NRC Lexicon	0.179^{*} (0.074)	0.160^{***} (0.009)
word_count	Other Controls	0.010^{**} (0.003)	0.023^{***} (0.000)
competitor_mention	Other Controls	-0.103 (0.426)	-0.746^{***} (0.046)
General UX Feedback	Topic Label	0.076 (0.252)	0.019 (0.025)
Music and Playlist Features	Topic Label	-0.717^{*} (0.288)	-0.395^{***} (0.025)
Other	Topic Label	-1.031^{***} (0.250)	-0.934^{***} (0.022)
Playback Issues	Topic Label	0.176 (0.229)	-0.148^{***} (0.024)
Positive Experience	Topic Label	-0.915^{**} (0.349)	-0.560^{***} (0.027)
Technical Fixes	Topic Label	-0.288 (0.255)	-1.098^{***} (0.027)
Observations		60,915	121,330
Log Likelihood		-1365.859	-74247.462
RMSE		0.06	0.46
McFadden R^2		0.044	0.117

Notes: Standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

NRC Lexicon variables are derived from the NRC Emotion Lexicon.

Topic Label reference category is **Account & Subscription**; updated for both models.

Coefficients represent changes in log-odds of receiving a Spotify reply.

Materials and Methods

Model Variables and Measurement. The logistic regression model predicting Spotify's reply behavior uses a combination of structured metadata and text-derived features. Key predictors include emotion scores from the NRC Lexicon (e.g., *disgust*, *joy*, *trust*), review word count, topic labels generated via LDA, and a binary indicator for competitor mentions. The outcome variable is whether the review received an official reply from Spotify.

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